

# **AWS maintenance / establishment checklist**

|                  |                     |            |                  |
|------------------|---------------------|------------|------------------|
| Station name     | ZACK H              |            |                  |
| People present   | ZACK H              |            | (Instructed by:) |
| Purpose of visit | MINOR MAINTENANCE   |            |                  |
| Date & time      | Arrival: 30.6 17:00 | Departure: |                  |
| Transport        | By: ARGO TRACKER    | From:      | ZACK STATION     |
| Weather          | SUNNY, WINDY        |            |                  |

**Remarks on state of AWS / maintenance / establishment**

AWS 015 Aug 2018

**Metadata upon arrival** (Establishment of new AWS: Go to last page)

|                                                                       |                        |             |
|-----------------------------------------------------------------------|------------------------|-------------|
| Time difference between logger clock and UTC                          |                        | + / -       |
| Adjust logger time to UTC (be sure PC runs on UTC time)               |                        | Y / (N)     |
| Download data to PC and/or change CF Card (wait for green light)      |                        | (Y) / N     |
| Name of logger program                                                |                        |             |
| Photo or screen dump of values in fast scan mode                      |                        | Y / (N)     |
| Latitude (dd mm.mmm)                                                  |                        |             |
| Longitude (dd mm.mmm)                                                 |                        |             |
| Altitude (m)                                                          |                        |             |
| Photos of tripod, sensors, logger box wiring etc. (more is better)    |                        | (Y) / N     |
| Mast tilt in boom direction (+ if radiometer tilting down)            |                        | °           |
| Mast tilt across boom looking from radiometer (- if clockwise)        |                        | °           |
| Boom direction relative to north                                      |                        | ° true/magn |
| Wind sensor box direction along boom? If not, measure direction.      |                        | Y / N       |
| Radiometer aligned with mast and inclinometer? If not, measure.       |                        | Y / N       |
| Temperature / humidity height (bottom of casing – surface)            | 2410                   | cm          |
| Sonic ranger on AWS height (membrane – surface)                       | 216                    | cm          |
| Wind sensor height (center of propeller – surface)                    | 2290                   | cm          |
| Sonic ranger on stakes height (membrane – surface)                    | fallen                 | cm          |
| Free length of stake – outer, holding boom with sonic ranger          |                        | cm          |
| Free length of stake – middle                                         |                        | cm          |
| Free length of stake – outer, holding boom without sonic ranger       |                        | cm          |
| Free length of ablation hose on/above the ice surface                 | frozen in              | cm          |
| Alt: depth of ablation hose using markings on hose                    | frozen in              | cm          |
| Height of ablation hose fluid level above the ice surface ~ 140       | frozen in              | cm          |
| Vertical difference in surface height (at station – at hose) (approx) | 9 ÷ 40                 | cm          |
| Length of thermistor string on surface from surface marking           | frozen in              | cm          |
| Height of surface irregularities                                      | 10                     | cm          |
| Stake assembly still standing?                                        | Y / (N) / more or less |             |
| Apparent damage to sensors? If yes, which?                            |                        | Y / N       |

Remarks:

station tilted down glacier and to high.  
Left due to loosened guy wires → tightened

### Serial numbers and maintenance details

| Part                                                             | Old part number                                                                                                                                        | New part number, if replaced | Potential maintenance tasks                                                                                                                                          |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Radiometer                                                       |                                                                                                                                                        |                              | - Aligned with mast: Y/N<br>- Clean: Y/N                                                                                                                             |
| Inclinometer                                                     |                                                                                                                                                        |                              | - Aligned with radiometer within 0.5°: Y/N                                                                                                                           |
| Satellite antenna                                                |                                                                                                                                                        |                              |                                                                                                                                                                      |
| Wind sensor                                                      |                                                                                                                                                        |                              |                                                                                                                                                                      |
| Temperature / humidity assembly (Fan, Bracket, MP100, Hygroclip) |                                                                                                                                                        |                              | - New HygroClip with old assembly: Y/N<br>- Fan spins and sounds OK: Y/N<br>Note Ass no. + serialno. for all 4 parts !                                               |
| Or: HygroClip only                                               |                                                                                                                                                        |                              | - Large offset in HygroClip temperature: Y/N                                                                                                                         |
| Sonic ranger on AWS                                              |                                                                                                                                                        |                              | - Old sensor with new membrane: Y/N<br>- Clicking as it should: Y/N                                                                                                  |
| Solar panel                                                      |                                                                                                                                                        |                              | - Output OK: Y/N<br>- Clean: Y/N                                                                                                                                     |
| GPS antenna                                                      |                                                                                                                                                        |                              | - Stuck to top of logger enclosure: Y/N                                                                                                                              |
| Data logger                                                      |                                                                                                                                                        |                              | - New internal battery in old logger: Y/N<br>- New operating system: Y/N<br>- New logger program: Y/N                                                                |
| Card reader                                                      |                                                                                                                                                        |                              | - LED either green or flashing orange (active): Y/N                                                                                                                  |
| Multiplexer                                                      |                                                                                                                                                        |                              | - Clicking as it should: Y/N                                                                                                                                         |
| Iridium modem IMEI number                                        |                                                                                                                                                        |                              | - Transmissions tested: Y/N<br>- Problems: Y/N                                                                                                                       |
| Barometer                                                        |                                                                                                                                                        |                              |                                                                                                                                                                      |
| Sonic ranger on stakes                                           |                                                                                                                                                        |                              | - Old sensor with new membrane: Y/N<br>- Clicking as it should: Y/N                                                                                                  |
| Thermistor string                                                |                                                                                                                                                        |                              | - New string in use, but old one still in ice: Y/N                                                                                                                   |
| Ablation hose assembly                                           |                                                                                                                                                        |                              | - New hose in use, but old one still in ice: Y/N<br>- Air in current hose: Y/N<br>- Air removed from current hose: Y/N<br>- Added 50% antifreeze mix to bladder: Y/N |
| Logger enclosure                                                 | - Replaced (including everything inside): Y/N<br>- New vent filter in logger enclosure: Y/N<br>- Moisture inside: Y/N<br>- New desiccant bags (2): Y/N |                              |                                                                                                                                                                      |
| Battery box                                                      | - New box with new batteries: Y/N<br>- Old box with new batteries: Y/N<br>- Moisture inside: Y/N<br>- New desiccant bags (2): Y/N                      |                              |                                                                                                                                                                      |
| Stakes                                                           | - New ones drilled in: Y/N<br>- Length: 2 / 4 / 6 / 8 / 10 m                                                                                           |                              |                                                                                                                                                                      |
| Tripod                                                           | - New parts: Y/N                                                                                                                                       |                              |                                                                                                                                                                      |

# **Metadata after maintenance / establishment**

|                                                                          |             |
|--------------------------------------------------------------------------|-------------|
| Time difference between logger clock and UTC                             | + / -       |
| Adjust logger time to UTC (be sure PC runs on UTC time)                  | Y / N       |
| Download data to PC and/or change CF Card (wait for green light)         | Y / N       |
| Name of logger program                                                   |             |
| Photo or screen dump of values in fast scan mode                         | Y / N       |
| Latitude (dd mm.mmm)                                                     |             |
| Longitude (dd mm.mmm)                                                    |             |
| Altitude (m)                                                             |             |
| Photos of tripod, sensors, logger box wiring etc. (more is better)       | Y / N       |
| Mast tilt in boom direction (+ if radiometer tilting down)               | °           |
| Mast tilt across boom looking from radiometer (- if clockwise)           | °           |
| Boom direction relative to north (radiometer should point to true south) | ° true/magn |
| Wind sensor box direction along boom? If not, change.                    | Y / N       |
| Radiometer aligned with mast and inclinometer? If not, align.            | Y / N       |
| Temperature / humidity height (bottom of casing – surface)               | cm          |
| Sonic ranger on AWS height (membrane – surface) 3503                     | cm          |
| Wind sensor height (center of propeller – surface)                       | cm          |
| Sonic ranger on stakes height (membrane – surface) 1811 22               | cm          |
| Free length of stake – outer, holding boom with sonic ranger             | cm          |
| Free length of stake – middle                                            | cm          |
| Free length of stake – outer, holding boom without sonic ranger          | cm          |
| Free length of ablation hose on/above the ice surface                    | cm          |
| Alt: depth of ablation hose using markings on hose                       | cm          |
| Height of ablation hose fluid level above the ice surface                | cm          |
| Vertical difference in surface height (at station – at hose) (approx)    | cm          |
| Length of thermistor string on surface from surface marking              | cm          |
| Height of surface irregularities                                         | cm          |

Remarks: